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RESEARCH ARTICLE

An Implicit Basis for the Retention Benefits of Random Practice

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ABSTRACT. The cognitive effort explanations of contextual interference (CI) and implicit motor learning represent a paradox in which cognitive involvement is seen to be advantageous or disadvantageous for learning. The authors aimed to resolve this paradox by measuring cognitive effort and working memory dependence during low and high CI practice on two Australian Rules Football tasks (kicking and handball). Measures of cognitive effort included: kicking and handball outcome performance during acquisition and during a test of retention, performance on a probe reaction time task during a sample of acquisition trials, and self-reported levels of cognitive effort. Measures of implicit and explicit learning included kicking and handball performance during a secondary task transfer, and self-report verbal protocols (number of verbal rules and hypotheses reported). The results suggest that high CI may cause an implicit mode of learning, perhaps due to the interference caused by task switching. However, these findings are restricted to the more complex of the 2 tasks (kicking).

Keywords: cognitive effort, contextual interference, explicit motor learning, implicit motor learning

Motor learning has been described as a problem-solving process in which the goal of an action represents a problem and the development of an appropriate movement pattern represents the solution (Guadagnoli & Lee, 2004). Skill acquisition researchers are interested in understanding which practice variables serve to optimize this process and consequently to understand the mechanisms that subserve these variables. The scheduling of practice trials (blocked and random practice) is one variable that has been shown to influence the efficacy of motor learning. Randomly sequencing practice trials throughout a training session (high contextual interference) has been shown to result in successful performance on tests of retention and transfer, despite poorer performance during acquisition. In contrast, repeating all practice trials for one task before switching to another (low contextual interference or blocked practice) has been shown to result in impaired performance on tests of retention and transfer, despite superior performance during acquisition. This effect is referred to as the contextual interference effect (for extensive reviews, see Brady, 1998, 2008; Lee & Simon, 2004; Magill & Hall, 1990).

Various researchers have proposed that the mechanism underlying the retention benefit of high-interference learning is the enhanced cognitive activity that may accompany random practice (Brady, 1998; Lee, Swinnen, & Serrien, 1994; Li & Wright, 2000; Rendell, Masters, & Farrow, 2009; Young, Cohen, & Husak, 1993). Others (e.g., Shea & Zimny, 1983) proposed that higher levels of cognitive effort occur for random learners during practice because they engage in better relational and or distinctive processing of their actions (i.e.,

they tend to compare and contrast the tasks that they are learning). As a result of this comparative processing, it is thought that random learners acquire more elaborate memory representations of the tasks that they have practiced. According to this account (referred to as the *elaboration and distinctiveness hypothesis*), the diminished performance of random learners during acquisition is caused by the need to actively differentiate the movement solutions of each task. The diminished effects of random practice are typically evident in experiments as longer latency values or larger error scores during acquisition (Del Ray, 1982; Shea & Morgan, 1979; Whitehurst & Del Ray, 1983; Wright, 1991; Wright, Li, & Whitacre, 1992).

In contrast, other researchers (e.g., Lee & Magill, 1983, 1985) considered the higher level of cognitive effort to be the result of demanding reconstructive processes that random learners must engage in when they switch between tasks. According to this explanation (referred to as the *action plan reconstruction hypothesis*), random learners forget the movement solution of each task as they process the task requirements of the other. On switching back to the initial task, they are forced to undergo a demanding reconstructive process to replan the way in which they perform the task. The need for random learners to repeatedly plan the movement solution results in poorer performance during acquisition, but ultimately promotes the retention of tasks because the learner is well practiced at reconstructing the motor solution (a factor that is helpful in tests of retention). Empirical evidence exists to support the reconstruction view (Gabriele, Hall, & Lee, 1989; Lee, Wishart, Cunningham, & Carnahan, 1997; Weeks, Lee, & Elliot, 1987).

Both the elaboration and reconstruction accounts of the contextual interference effect assume that motor skill learning is supported by a limited-capacity system, such as working memory (Baddeley, 2000; Baddeley & Hitch, 1974), which allows information to be temporarily stored and manipulated during the problem-solving process. The elaboration hypothesis suggests that task switching during random practice places a high level of demand on working memory resources due to the inclination to attend to more than one task in memory at a given moment. The reconstruction hypotheses suggests that task switching forces learners to repeatedly dump and then replan each movement solution in working memory as they alternate the allocation of their

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attentional resources between the different the skills they are learning (Lee & Simon, 2004).

Another area of research that has considered the relative importance of high and low demands on working memory during motor skill learning is implicit and explicit motor learning. According to Masters and Maxwell's (2004) model of motor skill acquisition (see also Liao & Masters, 2001; Masters, 1992; Maxwell, Masters, & Eves, 2003; Maxwell, Masters, Kerr, & Weedon, 2001), practice that makes few demands on working memory can result in specific learning benefits. Furthermore, according to this model, when learners do engage working memory resources during practice, they become susceptible to disruptions in movement control. This model suggests that explicit learning, which is characterized by the working memory-dependent process of generating movement strategies and evaluating their effectiveness, is likely to result in knowledge that is more susceptible to forgetting over time. Implicitly acquired knowledge on the other hand, is less dependent on working memory (Masters, MacMahon, & Pall, 2004; Maxwell, Masters, & Eves, 2000; Maxwell et al., 2003) and results in relatively less degradation over time (Reber, 1993).

It is clear that the cognitive effort explanations of the contextual interference effect and the findings from implicit motor learning research represent a paradox. One line of research suggests advantages of practice that involves working memory-dependent processes (i.e., the elaboration and reconstruction hypotheses for the contextual interference effect), whereas another line of research suggests advantages of practice that minimizes or circumvents the role of working memory (i.e., the implicit motor learning effect).

Previous research has empirically established that random practice is associated with higher levels of cognitive activity than blocked practice (Li & Wright, 2000). Research evidence to date, however, has not established that this increase in processing is necessarily task related. One explanation of the cognitive effort paradox is that the retention benefits of random practice are subserved by an implicit mode of learning. Specifically, it is possible that the working memory resources of random learners may be so overwhelmed by the information required to generate multiple motor solutions (i.e., task switching) that they are unable to test hypotheses and store rules or knowledge about the movement solutions that they generate. This article represents the first attempt to explore this proposition.

One measurement technique that aims to quantify the demands on attentional resources used during motor performance is probe reaction time (PRT). This method has recently been used in both the contextual interference and implicit learning research domains (Lam, Maxwell, & Masters, 2009; Li & Wright, 2000). The PRT methodology involves the addition of a discrete secondary task during the intertrial interval of the primary task of interest. The secondary task typically takes the form of an auditory tone to which participants are required to respond. The latency of the reaction time to the tone is taken as an indication of the

amount of attentional capacity that is being occupied at that moment in time to control performance of the primary task (Abernethy, 1988). The greater the attentional requirements of the primary task at that moment in time, the slower the reaction time to the secondary task.

A similar, though distinct, measure that has been used in the implicit motor learning research is secondary task transfer. This measure differs from PRT in that it employs a continuous secondary task to measure attentional control over a number of practice repetitions and is implemented following the learning phase. In this measure, participants perform the primary and secondary tasks under two conditions (independently and concurrently). Participants are typically instructed to give attentional priority to the primary task. Changes in performance on the primary or secondary task from single- to dual-task conditions are taken as an indication of how much attention the performer requires to conduct the primary task (Abernethy, 1988). It is thought that when processing demands exceed the available capacity, there is a reduction in performance on either the primary or secondary task (depending on the instructions given to participants regarding where they should allocate their attentional priority).

Both the PRT measure and the secondary task transfer measure are based on the limited capacity theory of attention. According to this theory, attention has a finite capacity that may be subdivided among tasks so long as the sum of attentional demand does not exceed the available capacity (Kahneman, 1973). Thus, when a primary task is characterized by a high processing demand, then there is only a small amount of residual capacity available to allocate to concurrent tasks, and thus secondary task performance is poor. In contrast, when processing demands are low, then there is a large amount of residual processing capacity available to allocate to concurrent tasks and thus secondary task performance is relatively good (Abernethy, 1988).

In the present study, we used the PRT and secondary task transfer measures within a battery of tests that also includes outcome accuracy, self-report ratings of cognitive effort, and verbal reports of explicit knowledge. We hypothesized that, relative to blocked practice, random practice would result in poor skill performance during acquisition, superior performance on tests of retention, and high levels of working memory demand during acquisition, as shown by poor performance on the PRT test and self-reports of high cognitive effort. Based on the suggestion that the cognitive effort experienced by random learners results in an implicit mode of learning, we further hypothesized that random practice would result in robust performance under secondary task transfer and the formation of minimal task relevant knowledge (reflected by sparse verbal reports).

Method

Participants

Nineteen participants (8 women, 11 men) with limited experience in playing Australian Rules Football (ARF) took

part in the study. Participants (M age = 28.5 years, SD = 7.09 years) possessed no more than 2 years experience in ARF over their lifetime, had not participated at a level higher than club, and none were involved in organized ARF activities in the period immediately preceding or during the study. Participants agreed not to practice the tasks or any skills related to ARF other than during the prescribed training sessions throughout the duration of the study. All participants provided informed consent before commencing the experiment.

Apparatus and Tasks

Two tasks, the drop punt kick and handball, were learned. The drop punt is a kicking technique that is the primary method of passing the ball between teammates in ARF. It is preferred by players over other methods of kicking because it is accurate and has a high speed of execution, which is important in the time-stressed game environment. The drop punt kick requires the player to hold the ball vertically and drop and kick it before it hits the ground, resulting in the ball spinning backwards end over end (Australian Football League, 2004). The handball task is an alternative method of passing the ball between teammates over relatively short distances. The handball requires the player to grip the ball with one hand and then hit it with a clenched fist using the other arm (referred to as the *punching arm*). A side-on stance is preferred to allow the punching arm to swing through freely (Australian Football League).

Previous researchers (e.g., Barreiros, Figueiredo, & Godinho, 2007) suggested that there is a need to bridge the gap between laboratory and applied research. Specifically, the use of unusual laboratory tasks and single-session acquisition phases incorporating massed practice conditions and short retention intervals have been highlighted as being of concern. Researchers have traditionally included deliberately exotic and novel tasks in studies of contextual interference (Barreiros et al.). The rationale behind the inclusion of the kicking and handball tasks in the present study was to incorporate real-world tasks to ensure that any findings could be generalized to real-life situations. Both of the tasks were modified to minimize the chance that the participant had previously encountered the particular movement actions that were required (i.e., to ensure that the tasks were novel to the participants). Although the kick roughly represented a drop punt kick, participants were constrained by a 4.5 m high roof which forced them to adopt more of a stabbing motion and use less follow through than they would in a regular drop punt kick. For the handball task, participants were required to use their nonpreferred arm as their punching arm.

The target for the kicking task (Figure 1) comprised a grid of 49 squares with sides of 50 cm in length (overall size = 3.5 m²). The center of the target was 1.75 m from the ground. The target was painted onto fabric, attached to a net, and positioned with the bottom edge level with the ground. Participants were instructed to aim for the central square of

1	2.36	3.29	3.63	3.29	2.36	1
2.36	4.01	5.25	5.75	5.25	4.01	2.36
3.29	5.25	6.99	7.88	6.99	5.25	3.29
3.63	5.75	7.88	10	7.88	5.75	3.63
3.29	5.25	6.99	7.88	6.99	5.25	3.29
2.36	4.01	5.25	5.75	5.25	4.01	2.36
1	2.36	3.29	3.63	3.29	2.36	1

FIGURE 1. Target and scoring values for the kicking task.

the target. Participants kicked from a distance of 15 m. This distance was chosen because it is the minimum distance that the ball must travel for the receiving player to claim a mark in the game of ARF (meaning that if the receiving player catches a ball that has been kicked more than 15 m, then the game stops while he prepares to kick).

The target for the handball task (Figure 2) consisted of six concentric rings painted onto fabric. The diameter of the center circle measured 30 cm and each additional ring increased by increments of 30 cm. The target was attached to a net and positioned with the centre at a height of 1.2 m (the lowest point of the outside ring was therefore 30 cm above the ground). Participants were instructed to aim for the central circle of the target. Participants handballed from a distance of 5 m.

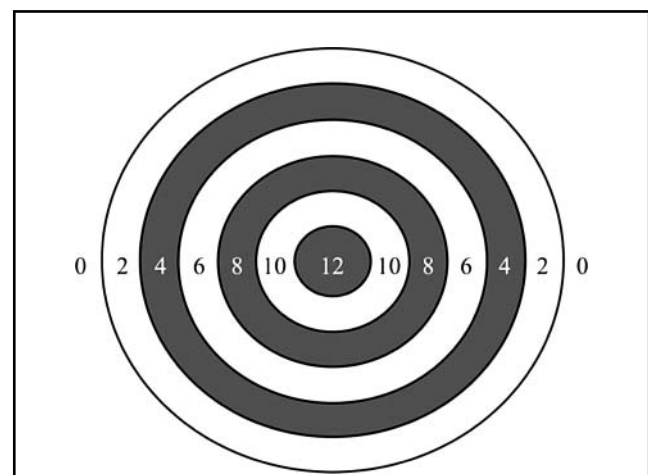


FIGURE 2. Target and scoring values for the handball task.

Measures

Performance Outcome (Accuracy)

Performance on the kicking and handball tasks was recorded by assigning scores to the target grids (Figures 1 and 2) and recording the outcome location of the ball as it hit the target. The experimenter recorded the scores in real time. Recordings from two video cameras that were focused on the targets were used to confirm reliability of the in situ scoring (100 trials were sampled to compare the in situ and video-based scoring methods: concurrence = 97%).

Probe Reaction Time

A probe reaction time (PRT) measure was used to assess the level of cognitive effort imposed by the kicking and handball tasks. Simple verbal reaction time to an auditory tone was recorded during intertrial intervals in the acquisition phase (see Procedure for more detail). Participants were instructed to respond by saying “tone” as quickly as possible upon presentation of the auditory tone. Probes were presented and recorded (in ms) with custom designed software (React) and a wireless lapel-mounted microphone that transmitted to a laptop computer.

National Aeronautics and Space Administration–Task Load Index

Four of the six dimensions of the National Aeronautics and Space Administration–Task Load Index (NASA-TLX) Questionnaire (Hart & Staveland, 1988) were used as a subjective measure of each participant’s cognitive effort. The four dimensions were mental demand, performance, effort, and frustration. In reference to mental demand, participants were asked, “How much mental and perceptual activity was required (e.g., thinking, deciding, calculating, remembering, looking, searching, etc.)? Was the task easy or demanding, simple or complex, exacting or forgiving?” For the performance dimension, participants were asked, “How successful do you think you were in accomplishing the goals of the task set by the experimenter (or yourself)? How satisfied were you with your performance in accomplishing these goals?” The question relating to effort was, “How hard did you have to work (mentally and physically) to accomplish your level of performance?” Finally, in terms of frustration, participants were asked, “How insecure, discouraged, irritated, stressed and annoyed versus secure, gratified, content, relaxed and complacent did you feel during the task?” The end-points for scales for mental demand, effort, and frustration were “low” and “high,” and for performance the endpoints were “good” and “poor.”

Secondary Task Transfer Test

Participants were asked to perform a secondary task while concurrently performing the kicking and handball tasks. High- (660 Hz) and low-pitched (440 Hz) tones were played to participants through computer speakers. Participants were required to indicate detection of the high

tones as rapidly as possible by saying “tone.” The high tones occurred randomly only 25% of the time. Tones were 500 ms in length and occurred once within every 2 s. This test has been shown to differentiate implicit learning from explicit learning, with implicit learners showing less disrupted motor performance when carrying out the secondary task (e.g., Poolton, Masters, & Maxwell, 2005). Participants were instructed to give equal priority to the primary and secondary tasks. Changes to primary task performance were assessed as an indication of attentional load.

Verbal Reports

Participants were asked to describe movements, methods or techniques that they had used consciously while performing the tasks. For each task (kicking, handball), the reports were collected following the pretest and the final acquisition session.

Design and Procedure

The study consisted of a pretest, acquisition period, transfer test, and retention test. All sessions occurred in a large indoor laboratory. Participants signed an informed consent form and provided information about demographics and previous experience. Three participants in each group had previously played the game for an average of 2 years, while the remaining participants had no previous playing experience.

Pretest

Participants were shown two instructional videos (each of approximately 5 min duration) that demonstrated the technique required to perform a drop punt kick and a handball. The videos were edited samples from a DVD entitled “Great Skills, Great Players” (Australian Football League, 2002). The sound was muted during the videos to prevent the participants from receiving verbal instructions about the tasks. Participants were instructed to duplicate the technique shown in the videos, but to adapt the technique to meet the demands imposed by the low roof height (in the kicking task) and the use of their nonpreferred hand (in the handball task). After viewing the videos, the participants performed a blocked schedule pretest of the kicking and handball tasks (trials = 20; 10 kicks, 10 handballs). Immediately following completion of the pretest, participants provided verbal reports and completed the NASA-TLX.

Acquisition Phase

Prior to the first acquisition session, participants were assigned to one of two groups, which differed in the practice schedule used: blocked ($n = 9$) and random ($n = 10$). The number of females and males were approximately equal in each group (blocked: females = 4, males = 5; random: females = 4, males = 6). The groups were matched for previous ARF experience. A t -test of the pretest accuracy scores confirmed that there were no significant performance differences

between the groups for the handball task, $t(17) = -0.05, p = .962$, or the kicking task, $t(17) = 0.13, p = .895$.

The acquisition phase consisted of 320 practice trials of both the kicking and handball tasks (640 trials per participant in total). The acquisition phase lasted 4 weeks, with two sessions per week for the first 3 weeks and one session in the final week. The first six training sessions included 50 practice trials of each task (100 trials total) and the final training session included 20 practice trials for each task (40 trials total). Fewer trials occurred in the final acquisition session because the participants were asked to also complete the transfer test within the same session. To include 100 practice trials within the session may have introduced issues of fatigue and boredom during the transfer test.

Within each session, the blocked learners practiced all trials for one task and then completed all trials for the other task. The order of tasks was counterbalanced across participants and alternated in each practice session. For the random group, participants practiced each task in a random order during each session. The random schedule was different for each practice session, but the same for each participant. The same task was not repeated for more than three consecutive trials in the random schedule.

Reaction-time probes were manually initiated by the experimenter to enable the probes to be presented during the intertrial interval (prior to movement initiation). Because the execution and intertrial interval was individual to each participant, it was not possible to initiate the probes at regular intervals. At the beginning of the first acquisition session, a baseline measure of participants' verbal reaction time was obtained from responses to 20 probe tones. The purpose of this baseline measure was to assess each individual participant's raw verbal reaction time (without the imposition of a concurrent task). For each consecutive block of 50 trials during the acquisition period, a different 10 trial segment was probed (see Figure 3). In the random condition, the kicking and handball trials were scheduled such that an equal number of kicking and handball trials occurred within the block of 10 trials that were probed with the auditory tone. No probes occurred in the final acquisition session due to the small number of trials completed. Participants

completed the NASA-TLX immediately upon completion of the acquisition trials in each session.

Secondary Task Transfer Test

The secondary task transfer test was conducted after a 10-min break following the final acquisition trial. The test incorporated a blocked schedule of 20 trials (10 kicks, 10 handballs) and required the participants to attempt to perform both the primary and the secondary tasks to the best of their ability (see *Measures* section for more details).

Retention Test

The retention test was conducted five weeks after the last acquisition session. Participants performed the kicking and handball tasks in a blocked schedule (trials = 20; 10 kicks, 10 handballs).

Analysis

Accuracy

Kicking accuracy was calculated using the scoring system outlined in Figure 1. A score of 10 was awarded for a ball that hit the central square of the target (the intended goal) and a score of 1 was awarded for a ball that hit the outermost corners of the target. A score of 0 was given if the ball missed the target area all together. Handball accuracy score was calculated using the scoring system outlined in Figure 2. A score of 12 was awarded if the ball hit the central circle of the target and a score of 2 was given if the ball hit the outermost ring on the target. A score of 0 was awarded for a ball that missed the target area. Acquisition trials during which there was a reaction time probe were excluded from analysis.

PRT

Mean verbal reaction time to the probes (in ms) was calculated separately for the kicking and handball tasks. Reaction times that were greater than two standard deviations above or below the mean were removed as outliers.

Session	Block of trials with probe on each trial									
	1st 10	2nd 10	3rd 10	4th 10	5th 10	6th 10	7th 10	8th 10	9th 10	10th 10
Acquisition 1										
Acquisition 2										
Acquisition 3										
Acquisition 4										
Acquisition 5										
Acquisition 6										
Acquisition 7										

FIGURE 3. Scheduling of probes within acquisition trials for the probe reaction time measure.

NASA-TLX

Each of the four dimensions was recorded on a 20-point bipolar scale. A score from 0 to 100 on each dimension was obtained by incrementing a score of 5 to each scale point.

Verbal Reports

Verbal protocols were scored by two independent raters, who then compared scores until a consensus was reached. The raters were blind to the experimental conditions under which each participant performed. The raters determined whether statements were related to movement (mechanical rule) or the testing of a hypothesis (hypotheses). The number of mechanical rules reported (e.g., handball: "I gripped the ball lightly with the platform hand"; kick: "I made contact with the ball on the bottom point") and the number of hypotheses tested (e.g., handball: "I adjusted the point of contact on the ball"; kick: "I tried to adjust my follow-through") were then summed. Any statements that were irrelevant to the technical demands of the task (e.g., "I enjoyed the session") were not included in the analysis because we were only interested in the rules and hypotheses that were specifically related to the tasks that participants were learning. A minimal number of hypotheses were reported by each group for both the kicking and handball tasks at pretest and after the last acquisition session (all $M_s < 0.7$), so mechanical rules and hypotheses were combined and an analysis was performed on the total amount of task relevant knowledge reported following the pretest and the last acquisition session.

Results

Performance During the Acquisition Period

We hypothesized that relative to blocked practice, random practice would result in poor skill performance during the acquisition period. To test this hypothesis, kicking and handball performance during the acquisition period were evaluated with separate Learning Group $\times$ Acquisition Session (2×7) analyses of variance (ANOVAs) with repeated measures on the acquisition session factor. There was a significant improvement in performance across the acquisition sessions for both the handball ($F(6, 102) = 7.76, p < .05$, partial $\eta^2 = .313$) and kicking tasks, ($F(6, 102) = 3.58, p < .05$, partial $\eta^2 = .174$). However, there was no significant main effect for learning group in either task: for handball, $F(1, 17) = 0.19, p = .669$; for kicking, $F(1, 17) = 0.03, p = .856$. There were also no significant learning group by acquisition session interaction effects for the handball, $F(6, 102) = 0.37, p = .897$, or kicking tasks, $F(6, 102) = 1.27, p = .279$. Figure 4 shows performance of the handball and kicking tasks.

Performance During the Retention Test

We hypothesized that relative to blocked practice, random practice would result in superior performance on tests of retention. To test this hypothesis, kicking and handball perfor-

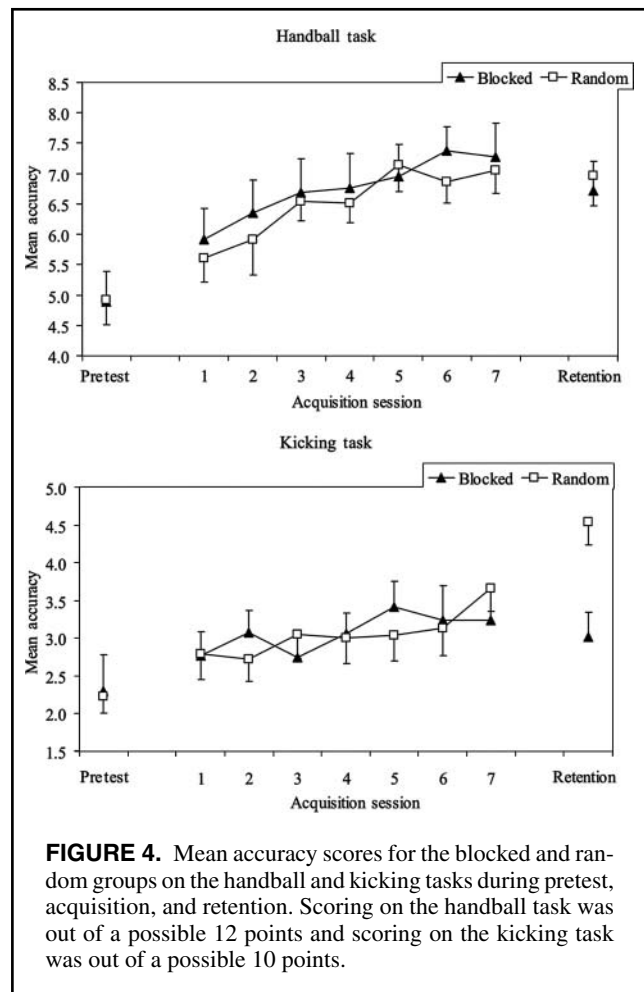


FIGURE 4. Mean accuracy scores for the blocked and random groups on the handball and kicking tasks during pretest, acquisition, and retention. Scoring on the handball task was out of a possible 12 points and scoring on the kicking task was out of a possible 10 points.

mance during the pretest and the retention test were analyzed with two separate Learning Group $\times$ Test Occasion (2×2) ANOVAs with repeated measures on the test occasion factor. For the handball task, there was a significant main effect for test occasion ($F(1, 17) = 33.69, p < .001$, partial $\eta^2 = .665$), such that the level of performance for both groups improved from the pretest to the retention test (see Figure 4). There was neither a significant main effect for learning group ($F(1, 17) = 0.06, p = .813$), nor a significant interaction effect ($F(1, 17) = 0.11, p = .747$).

For the kicking task, there was a significant main effect for test occasion ($F(1, 17) = 22.56, p < .05$, partial $\eta^2 = .570$); however, there was no significant main effect for learning group ($F(1, 17) = 3.64, p = .073$). There was a significant learning group by test occasion interaction ($F(1, 17) = 6.31, p < .05$, partial $\eta^2 = .271$). Paired sample t tests confirmed the impression given in Figure 4 that there were no significant differences from pretest to retention for the blocked group ($t(8) = -1.25, p = .248$), whereas the random group showed a significant improvement in performance from the pretest to the retention test in the kicking task ($t(9) = -7.28, p < .001$).

PRT

There were no significant differences between the blocked ($M = 474$ ms, $SD = 63$ ms) and random ($M = 466$ ms, $SD = 59$ ms) learners on the baseline measure of verbal reaction time ($t(17) = 0.267$, $p = .792$). We hypothesized that relative to blocked practice, random practice would result in higher levels of working memory demand during acquisition as shown by poor PRT performance. To test this hypothesis, PRT during the acquisition phase was analyzed using two separate Learning Group $\times$ Acquisition Session (2×6) ANOVAs with repeated measures on the acquisition session factor, and reaction time to the probe (handball or kicking) as the dependent variable.

For the handball task, the analysis revealed that there was a significant decrease in PRT across the acquisition sessions, ($F(5, 85) = 4.97$, $p < .001$, partial $\eta^2 = .226$), in the absence of a significant learning group effect ($F(1, 17) = 4.14$, $p = .058$), or an interaction effect ($F(6, 85) = 0.28$, $p = .921$). For the kicking task, the analysis revealed that the random learners were significantly slower than the blocked learners ($F(1, 17) = 4.94$, $p < .05$, partial $\eta^2 = .276$), in the absence of a significant effect for acquisition session ($F(5, 85) = 1.92$, $p = .097$), or an interaction effect ($F(5, 85) = 0.85$, $p = .517$). Figure 5 shows the mean PRT for blocked and random learners during the handball and kicking tasks.

NASA-TLX

The high levels of working memory demand hypothesized to occur during random practice were also expected to result in high self-report levels of cognitive effort (on the NASA-TLX). To test this hypothesis, four separate Learning Group $\times$ Test Occasion (2×7) ANOVAs were conducted with repeated measures on the test occasion factor and the subjective ratings of mental demand, effort, frustration, and performance as the dependent variables. The NASA-TLX was administered during each of the seven acquisition sessions. Three participants (2 from the blocked group and 1 from the random) group failed to respond to the questions on one of the seven testing occasions due to time constraints. All participants who had missing data were excluded from the NASA-TLX analysis. Given that these omissions were not on a systematic basis, they are unlikely to have an effect on the group data.

The random learners reported higher scores than the blocked learners on the measures of mental demand ($F(1, 14) = 5.94$, $p < .05$, partial $\eta^2 = .298$), effort ($F(1, 14) = 8.83$, $p < .05$, partial $\eta^2 = .387$), and frustration ($F(1, 14) = 4.62$, $p < .05$, partial $\eta^2 = .248$). Ratings also differed significantly across the test occasions for the measures of mental demand ($F(6, 84) = 9.28$, $p < .05$, partial $\eta^2 = .399$), effort ($F(6, 84) = 5.23$, $p < .05$, partial $\eta^2 = .272$), and frustration, ($F(6, 84) = 4.62$, $p < .05$, partial $\eta^2 = .48$). However, there were no significant interaction effects between learning group and test occasion for the measures of mental demand, effort, or

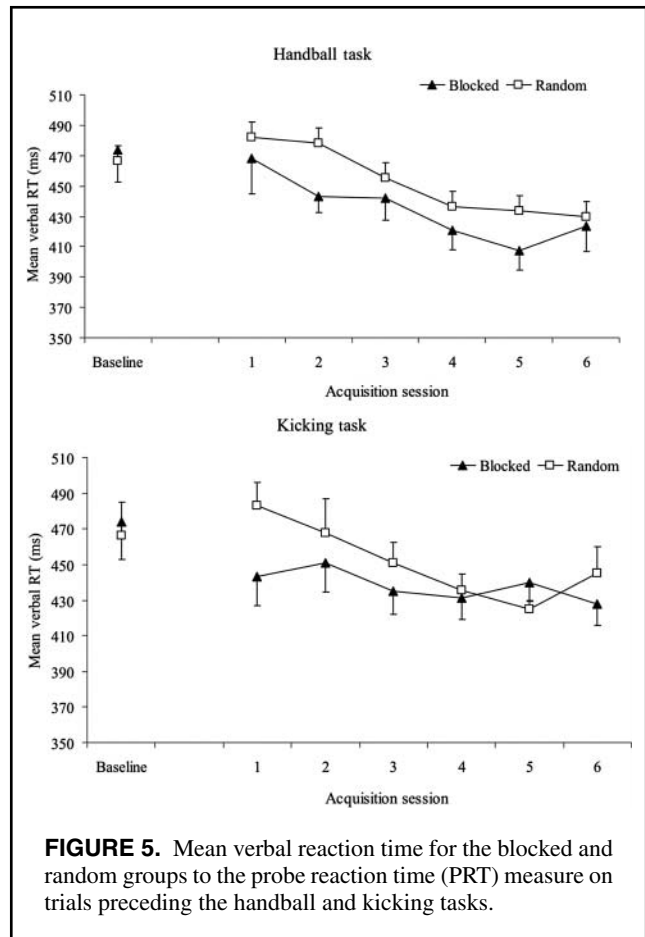


FIGURE 5. Mean verbal reaction time for the blocked and random groups to the probe reaction time (PRT) measure on trials preceding the handball and kicking tasks.

frustration: for (mental demand, $F(6, 84) = 2.00$, $p = .073$; for effort, $F(6, 84) = 0.57$, $p = .752$; and for frustration, $F(6, 84) = 1.09$, $p = .374$). For the performance variable, there were neither significant main effects for learning group ($F(1, 14) = 1.52$, $p = .238$) or test occasion ($F(6, 84) = 2.18$, $p = .053$), nor any significant interaction effects ($F(6, 84) = 0.45$, $p = .843$). Figure 6 shows the blocked and random groups' mean subjective ratings on the NSAS-TLX scales (Mental Demand, Effort, Frustration, and Performance).

Performance During the Transfer Test

Based on the suggestion that the cognitive effort experienced by random learners results in an implicit mode of learning, we hypothesized that random practice would result in robust performance under secondary task transfer. To test this hypothesis, paired sample t tests were used to compare the single- and dual-task performance of the blocked and random learners. For the handball task, there was no difference in performance (handball accuracy) from single- to dual-task conditions for either the blocked ($t(8) = 0.636$, $p = .542$) or random, ($t(9) = 0.506$, $p = .625$) groups (see Figure 7). For the kicking task, there was no difference in performance (kicking accuracy) from single- to dual-task conditions for the blocked group ($t(8) = -0.048$, $p = .963$); however,

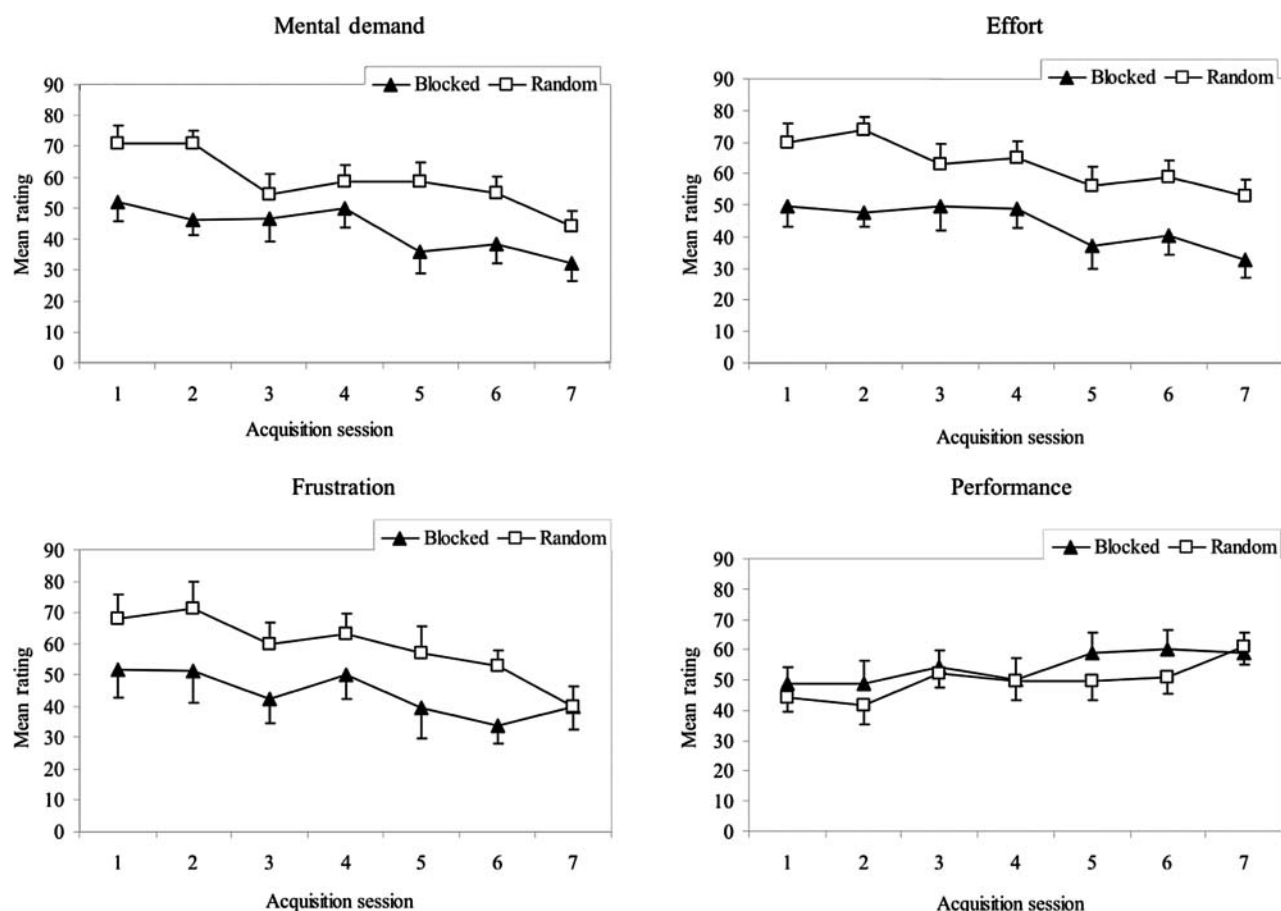


FIGURE 6. Mean subjective rating for the blocked and random groups on the mental demand, effort, frustration, and performance scales of the National Aeronautics and Space Administration–Task Load Index (NASA-TLX).

the random group showed a significant improvement from single- to dual-task conditions ($t(9) = -2.516$, $p < .05$; see Figure 7).

Verbal Reports

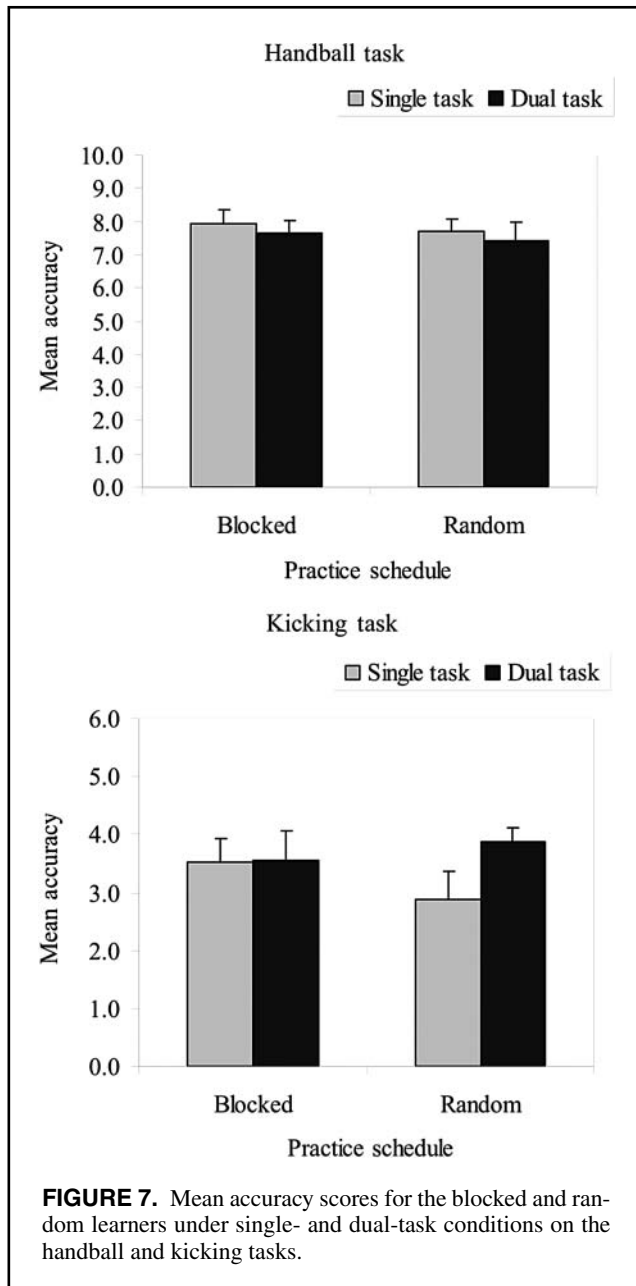
Our final hypothesis was that random learners would form minimal task relevant knowledge (reflected by sparse verbal reports) due to the implicit mode of learning for random learners. To test this hypothesis, paired sample t tests were used to compare the amount of task relevant knowledge reported by the blocked and random learners after the pretest session compared to the amount reported after the last acquisition session. For the handball task, there was no significant difference for either the blocked ($t(8) = 1.65$, $p = .137$) or random learners ($t(9) = 1.07$, $p = .313$; see Figure 8). For the kicking task, there was no difference for the blocked group ($t(8) = 1.05$, $p = .325$); however, there was a trend indicating that random group reported less task relevant knowledge following the acquisition period, although this finding did

not reach statistical significance ($t(9) = 2.20$, $p = .055$; see Figure 8).

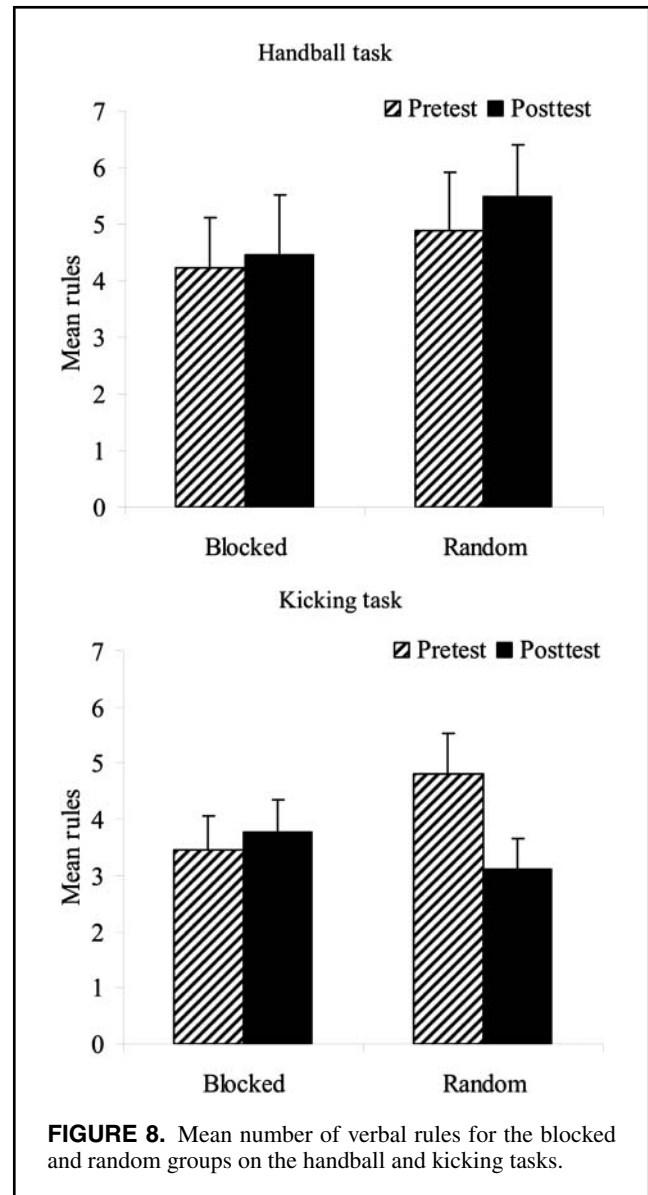
Discussion

The purpose of this study was to explore the hypothesis that the retention benefits of random practice are subserved by the mechanisms underlying implicit motor learning. We were interested in whether random learners would be overwhelmed by high levels of cognitive effort due to interference from task switching that would prevent them from consciously interpreting their movement outcomes (and consequently accruing explicit task-relevant knowledge). The results provide some support for this proposition; however, the effect appears to be mediated by the task being learned.

In line with previous research showing that random learners experience high levels of cognitive effort (e.g. Brady, 1998; Lee et al., 1994; Li & Wright, 2000; Young et al., 1993), we hypothesized that, relative to blocked practice, random practice would result in poor performance during acquisition, superior retention, slower PRTs during



acquisition, and high self-report levels of cognitive effort. The results delivered mixed findings regarding these hypotheses. First, contrary to expectations, there was no difference between the level of performance of the blocked and random learners during acquisition. This finding contradicts the traditional contextual interference effect, but supports research suggesting that high contextual interference schedules are less likely to adversely affect performance during acquisition of applied tasks than laboratory based tasks. For instance, Goode and Magill (1986) did not observe between-groups differences during acquisition in their applied study of blocked, random, and serial practice of badminton serves (despite finding significant differences between the learning groups on both the retention and transfer tests). Similarly, in



their comprehensive review of contextual interference studies in applied settings, Barreiros et al. (2007) found that the suppressed performance for random learners was not evident during the acquisition phase in the majority (61%) of studies that assessed acquisition performance.

With reference to retention, the findings were skill dependent. For the handball task, there were no retention benefits for the random practice schedule compared to the blocked schedule. In contrast, for the kicking task, there was a significant interaction between the groups, demonstrating that the processes subserving kick execution adapted differentially as a consequence of random compared to blocked learning. The random learners were remarkably successful on the retention test, indicating that the learning benefits of random practice were evident for the kicking task in the present study. The degree of variation in the skills used in contextual

interference studies has been highlighted in the existing literature. Magill and Hall (1990) suggested that including skills from different motor programs is likely to increase the amount of interference caused during the learning process (this is based on Schmidt's [1975, 1988] view of a motor program). Magill and Hall suggested that when tasks were consistent on aspects such as relative timing, sequence of events, and spatial configurations, then the tasks were unlikely to introduce a sufficient level of interference to produce the traditional interaction effect. This was the basis for our inclusion of two distinctly different skills (kicking and handball). Given that we observed significantly disparate trends for our two tasks in relation to retention, it may have been preferable (in terms of gaining clear results) to select skills which fulfilled the criteria of being from two different motor programs, but which were of the same classification of skills (e.g., two different badminton serves, as adopted by Goode and Magill [1986]; driving, middle distance swing, pitching, and chipping in golf, as adopted by Brady [1997]). This suggestion is echoed by the challenge point framework advocated by Guadagnoli and Lee (2004), which we consider in more detail subsequently.

The task being learned also appeared to differentiate probe reaction time responses in the different schedules. There was no difference between the blocked and random learners in PRT performance during the handball task; however, the random learners displayed significantly slower PRTs than the blocked learners during the kicking task. This finding supports the contention that random learners experienced higher levels of cognitive effort during practice, but in the more demanding kicking task only. This finding also highlights the importance of task differences that we discussed previously in relation to the retention findings.

In light of the observation that performance at retention and on the PRT measure was differentially affected by task complexity, it is unfortunate that we did not separate ratings for the kicking and handball tasks when we administered the self-report measure of cognitive effort. Overall, however, the results from this measure suggest that random learners experienced higher levels of mental demand, effort, and frustration during the acquisition phase of the study. These findings provide further support for our hypothesis that random learners experience higher levels of cognitive effort than do blocked learners. The findings are also consistent with previous studies that highlighted the role of cognitive effort in contextual interference studies (Brady, 1998; Lee et al., 1994; Li & Wright, 2000; Young et al., 1993).

The weight of the evidence (from the retention data, PRT measure, and self-report ratings) suggests that the level of cognitive effort was higher for the random learners than for the blocked learners (for the more difficult of the two tasks: the kicking task). The data relating to the second set of hypotheses also showed differential results for the kicking and handball tasks. Based on the suggestion that the cognitive effort experienced by random learners results in an implicit mode of learning, we hypothesized that random

practice would result in robust performance on a secondary task transfer test, and the accrual of minimal verbal rules and the testing of few hypotheses.

The introduction of cognitive load in the secondary task transfer test provides a measure of the working memory resources required to perform the primary task. With reference to the first primary task (handball), there were no significant differences in handball accuracy for the blocked and random learners between the single- and dual-task conditions. This finding demonstrates that both groups were able to cope with the demands of the secondary task. The results for the second primary task (kicking) were disparate to those of the handball task. For the blocked learners, there was no significant difference in performance on the kicking task between the single- and dual-task conditions. However, for the random learners, there was a significant improvement in performance on the kicking task under secondary task load.

One interpretation of the finding that the random learners were able to perform the kicking task exceptionally well under the demands of a secondary task is that they were able to perform the kicking task with minimal demands on working memory. This finding is consistent with prior studies of implicit motor learning (e.g., Masters, 1992; Masters, Poolton, Maxwell, & Raab, 2008; Maxwell et al., 2003) and also with studies showing that the addition of a secondary task can be beneficial to primary task performance because it reduces online attentional control (Beilock, Carr, MacMahon, & Starkes, 2002). Implicit and explicit processes occur on a continuum rather than being two distinct categories of learning. This means that the level of performance achieved under secondary task load reflects the amount of processing capacity required to perform the primary task (and hence gives an indication of the level of implicit control). Superior performances under secondary task load can be interpreted as relying on a lesser amount attentional control (and hence are more implicit), whereas poorer performances under secondary task load can be interpreted as relying on a greater amount of attentional control (and hence are more explicit). It is possible that maintaining performance under secondary task load indicates some degree of implicit control (as was the case in the blocked condition for both skills and the random condition for the handball skill). However, we suggest that the random group's exceptional performance on the kicking task under secondary task load indicated that the level of reliance on implicit processes was greater for in this condition than in any other testing condition in the current study.

Successful performance on the secondary task used in the present study (verbally identifying high-pitched tones from an array of high- and low-pitched tones) is dependent on a number of psychological processes, such as encoding, matching, and response processes. However, it is arguable that this secondary task is not as difficult as those used in previous implicit learning studies, such as random letter generation (Poolton, Masters, & Maxwell, 2007) and counting backwards (Liao & Masters, 2001), which call on the short-term

storage or rehearsal resources of memory. This is a possible reason why there were no decrements in performance under secondary task load for the blocked learners in either task or the random learners in the handball task.

An alternative interpretation of the secondary task transfer results is that the exceptional performance for the random group on the kicking task under secondary task load was due to a decrement in performance under single-task conditions rather than an improvement under dual-task conditions *per se*. If this interpretation is correct, then the reason is likely to be that the secondary task transfer testing was done under blocked conditions. The disruption in single-task performance for the random group would be due to the change in processing that occurred as they switched from a random schedule during acquisition to a blocked schedule on the single-task transfer test. Specifically, the nature of the blocked trials might have enabled the random learners to begin hypothesis testing based on the immediate feedback from their previous trial and hence cause them to adopt a more explicit mode of control.

The final measure explored in this study was the amount of task-relevant knowledge reported by participants. We looked at the amount of knowledge that was reported following a blocked schedule pretest compared to the amount reported immediately after both groups had completed their final acquisition session using the different practice schedules. We were interested in whether the random group reported fewer rules and hypotheses following learning because this would indicate that they had less task-relevant knowledge available after random practice. It would also suggest that they had adopted a more implicit mode of control similar to the type of learning seen in other paradigms, such as loading working memory with a demanding secondary task (dual-task model: Hardy, Mullen, & Jones, 1996; MacMahon & Masters, 2002; Masters, 1992); explaining the skill requirements by analogy or metaphor (analogy learning: Law, Masters, Bray, Eves, & Bardswell, 2003; Liao & Masters, 2001); withholding visual and auditory information from the learner (reduced feedback learning: Masters, 2000; Maxwell et al., 2000); giving task-related but goal-irrelevant instructions (Farrow & Abernethy, 2002); and simplifying task demands to prevent errors and hypothesis testing (errorless learning: Maxwell et al., 2001; Poolton et al., 2005). As was the case with the majority of measures in this study, the results were different for the kicking and handball tasks. For the handball tasks, both the blocked and random learners did not differ from pretest to posttest in the amount of task relevant knowledge reported. However, for the kicking task, the random group tended to report a smaller amount of task relevant knowledge when they were assessed following the final acquisition. Although this finding did not reach statistical significance, it suggests that it is possible that the random group acquired the kicking task using similar implicit processes to those used in other implicit learning paradigms. The blocked group showed no change in task knowledge following the intervention.

The findings from the secondary task transfer test and the verbal reports point toward the possibility that the random learners adopted, or were forced to adopt, an implicit mode of control during acquisition (for the kicking task only). These findings again highlight the proposition that the task being learned is a critical factor in the interpretation of contextual interference results, a suggestion that complements the challenge point framework (Guadagnoli & Lee, 2004). In their model, Guadagnoli and Lee differentiated two types of task difficulty: nominal task difficulty, which reflects a constant amount of task difficulty regardless of task level or conditions, and functional task difficulty, which refers to how challenging the task is relative to skill level and the conditions experienced. The ideas presented within the challenge point framework are constructed as a series of predictions and empirical data is required to verify whether these predictions hold true. One prediction of the challenge point framework is that the advantage of random practice for learning is largest for tasks of the lowest nominal task difficulty. Although the present study was not a direct test of the challenge point framework, the finding that the advantage of random practice was largest for the kicking task contradicts this prediction of the challenge point framework. It is arguable that the kicking task has a higher nominal task difficulty than the handball task because (a) it requires a higher number of degrees of freedom, (b) it involves interception with a moving object (the ball was dropped before it was kicked, whereas in the handball, the platform hand supported the ball while the opposite arm swung through to contact it), and (c) because the release point of the ball was considerably further away from the target (kicking task: 15 m; handball task: 5 m).

This study represents a systematic attempt to quantify the type of cognitive processing associated with different practice schedules using a number of existing measures. The findings complement our suggestion that random practice may be subserved by the same mechanisms as those underlying implicit motor learning. Further research is required to verify these findings and future researchers should attempt to systematically manipulate the practice environment to help understand the contradiction between the two theories explored here. One possible methodological approach may be to observe the consequences of preventing elaboration or reconstruction in random practice (e.g., by using a cognitively demanding secondary task). If the retention benefits of random practice are dispelled in a condition where cognitive effort is prevented, then perhaps we have further support for the existing theories of contextual interference. If, on the other hand, the retention benefits of random practice are maintained, alternative explanations of the contextual interference effect, such as an implicit motor learning explanation warrant further consideration.

In summary, we have investigated the contextual interference effect using a number of measures that have typically been applied to implicit motor learning research. Our aim was to highlight the discrepancy between research outcomes from those two research areas and to provide some empirical

data that helps to gain insight to the cognitive effort paradox. Overall, our results lend support to the notion that the cognitive effort experienced by random learners results in an implicit mode of learning, perhaps due to the interference caused by task switching. However, these results were based on the outcomes of the more complex of the two learned tasks (the kicking task), which highlights the importance of task difficulty in the interpretation of the contextual interference effect. Further research involving systematic manipulations of cognitive effort in the contextual interference and implicit motor learning domains are needed to fully understand the mechanisms underlying the contextual interference effect.

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